Demo_6_FStatisticsInLinearModels_2025

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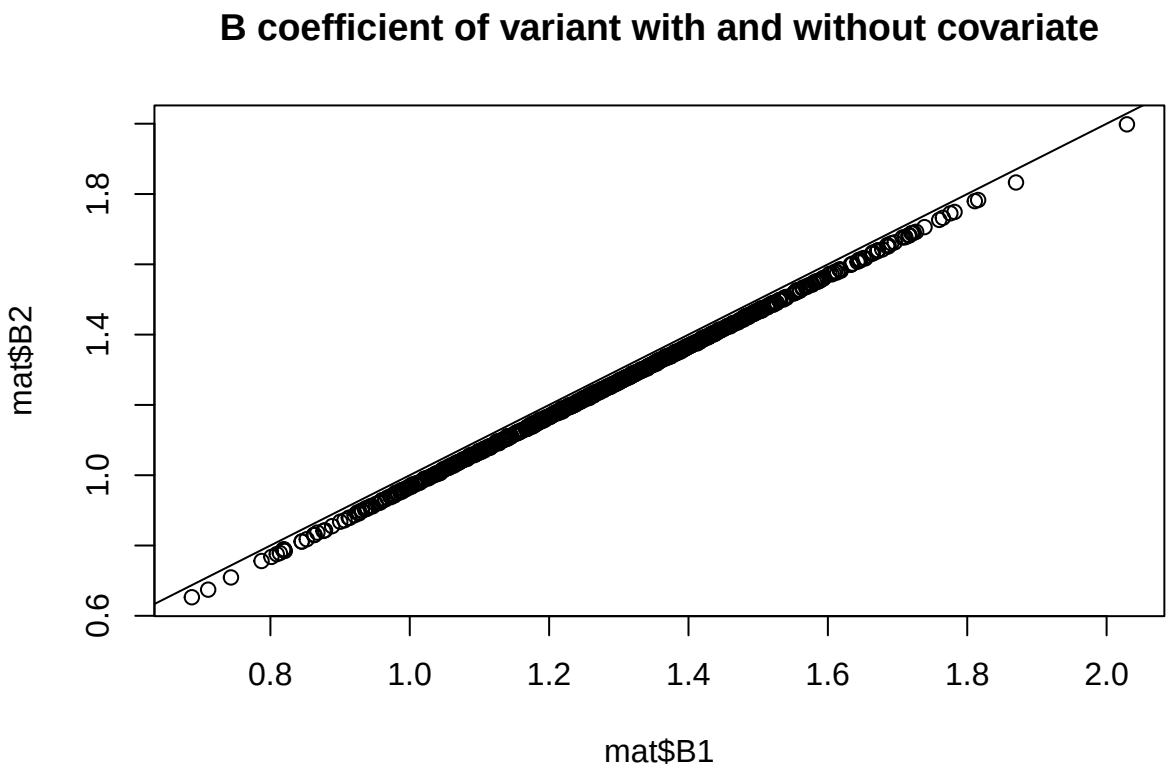
```r
knitr::opts_chunk$set(echo = FALSE)
knitr::opts_chunk$set(message = FALSE)
knitr::opts_chunk$set(warning = FALSE)

library(tinytex)

## Simulate Genetic Data for Linear Regression
### Define Parameters and Simulate Data
### Number of iterations
Iters <- 1000

### Define a matrix to store output summary statistics
mat <- matrix(rep(0, 5 * Iters), nrow = Iters, ncol = 9)
```

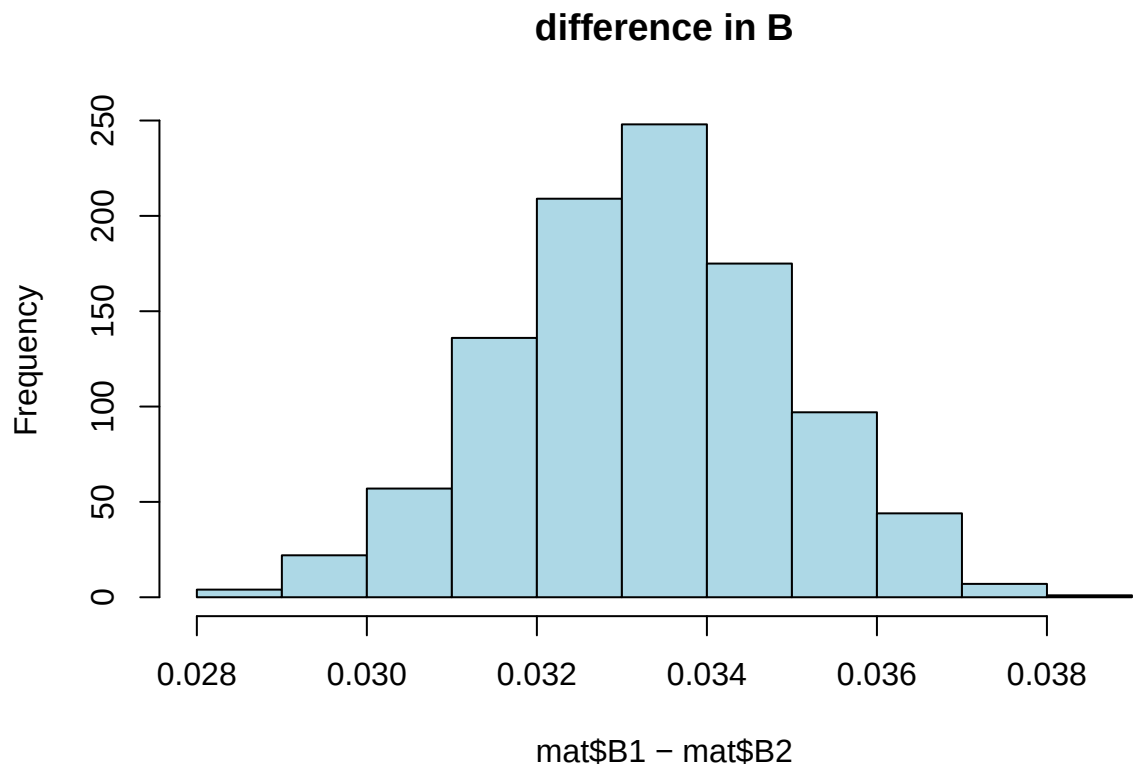
```
## Warning in matrix(rep(0, 5 * Iters), nrow = Iters, ncol = 9): data length
## [5000] is not a sub-multiple or multiple of the number of columns [9]
```

```r
### Effect allele frequency
q <- 0.3

### Number of observations
N <- 1000

### Complementary allele frequency
p <- 1 - q

### Effect size for heterozygotes (assume additive model for now)
B1 <- 1.25

### Effect size for another covariable
B2 <- 5

### Define allele counts from frequencies based on Hardy-Weinberg Equilibrium (HWE)
### What if we didn't round?
Nrr <- round(p^2 * N)
Nra <- round(2 * p * q * N)
Naa <- N - Nrr - Nra

### Define genotype vectors
Xrr <- rep(0, Nrr)
Xra <- rep(1, Nra)
```

```r
Xaa <- rep(2, Naa)
X <- c(Xrr, Xra, Xaa)

### Define covariable vector
Z<-rbinom(N,1,.5)

## Simulate Phenotypes and Perform Linear Regression

for (i in 1:Iters) {
  ### Simulate phenotypes (change lm to glm for binary phenotypes)
  Y <- (B1 * X) + (B2 * Z) + rnorm(N, 10, 4) #note that we apply 'noise' here using arbitrary parameter

  ### Fit linear model
  model <- lm(Y ~ X)

  ### Extract p-value and effect size of X
  p <- summary(model)$coefficients[2, 4]
  B <- summary(model)$coefficients[2, 1]
  f_statistic <- summary(model)$fstatistic[1]
  f_statistic_p <- pf(f_statistic, summary(model)$fstatistic[2], summary(model)$fstatistic[3], lower.ta

  ### Store iteration number, p-value and effect size for X
  mat[i, 1] <- i
  mat[i, 2] <- p
  mat[i, 3] <- B
  mat[i, 4] <-f_statistic
  mat[i, 5] <-f_statistic_p

  ### Fit a new model with the same data, this time including independent covariable Z
  model <- lm(Y ~ X + Z)

  ### Extract p-value and effect size.
  p <- summary(model)$coefficients[2, 4]
  B <- summary(model)$coefficients[2, 1]
  f_statistic <- summary(model)$fstatistic[1]
  f_statistic_p <- pf(f_statistic, summary(model)$fstatistic[2], summary(model)$fstatistic[3], lower.ta
  ### Store the p-value and effect size for X in the other model
  mat[i, 6] <- p
  mat[i, 7] <- B
  mat[i, 8] <-f_statistic
  mat[i, 9] <-f_statistic_p

}

### Convert matrix to data frame and name columns
mat <- as.data.frame(mat)
colnames(mat) <- c("iter", "p1", "B1","F1","F1p","p2", "B2","F2","F2p")

plot(mat$B1,mat$B2,main="B coefficient of variant with and without covariate")
abline(0,1)
```




```
hist(mat$B1-mat$B2,breaks=10,col="light blue",main="difference in B")
```



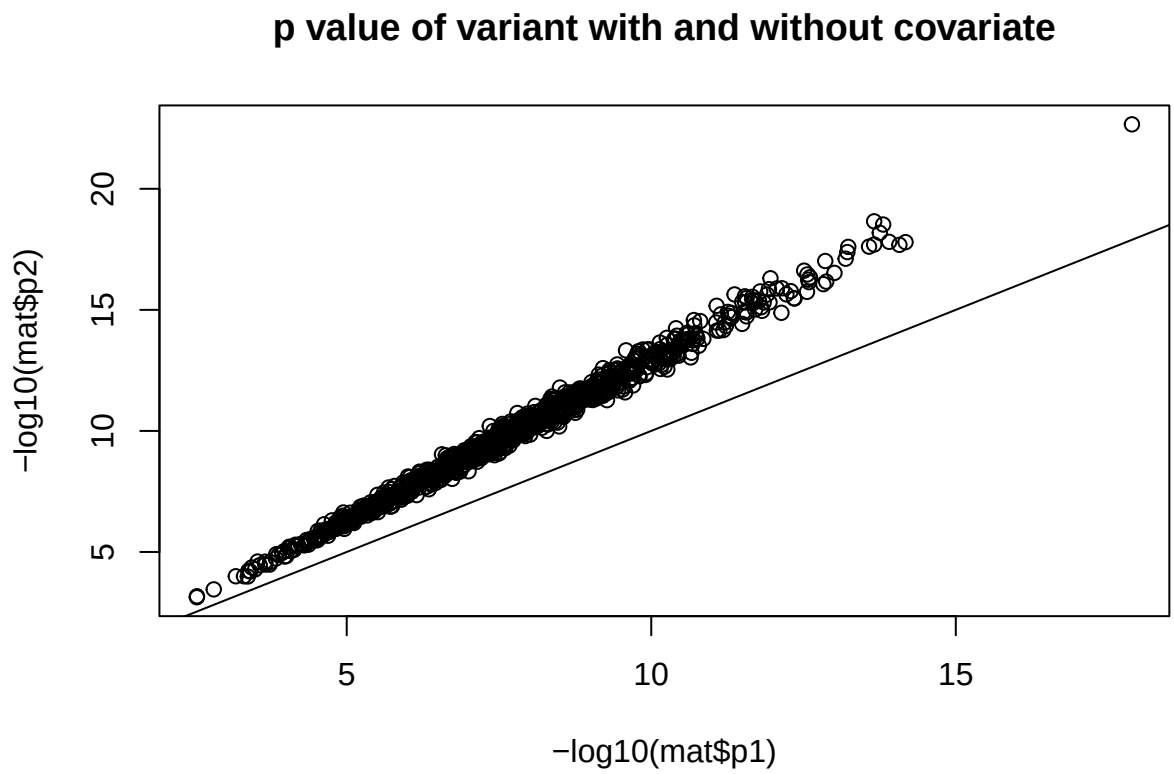

```r
mean(mat$B1-mat$B2)
```

```
## [1] 0.03327996
```

```r
max(mat$B1-mat$B2)
```

```
## [1] 0.03804603
```

```r
plot(-log10(mat$p1),-log10(mat$p2),main="p value of variant with and without covariate")
abline(0,1)
```



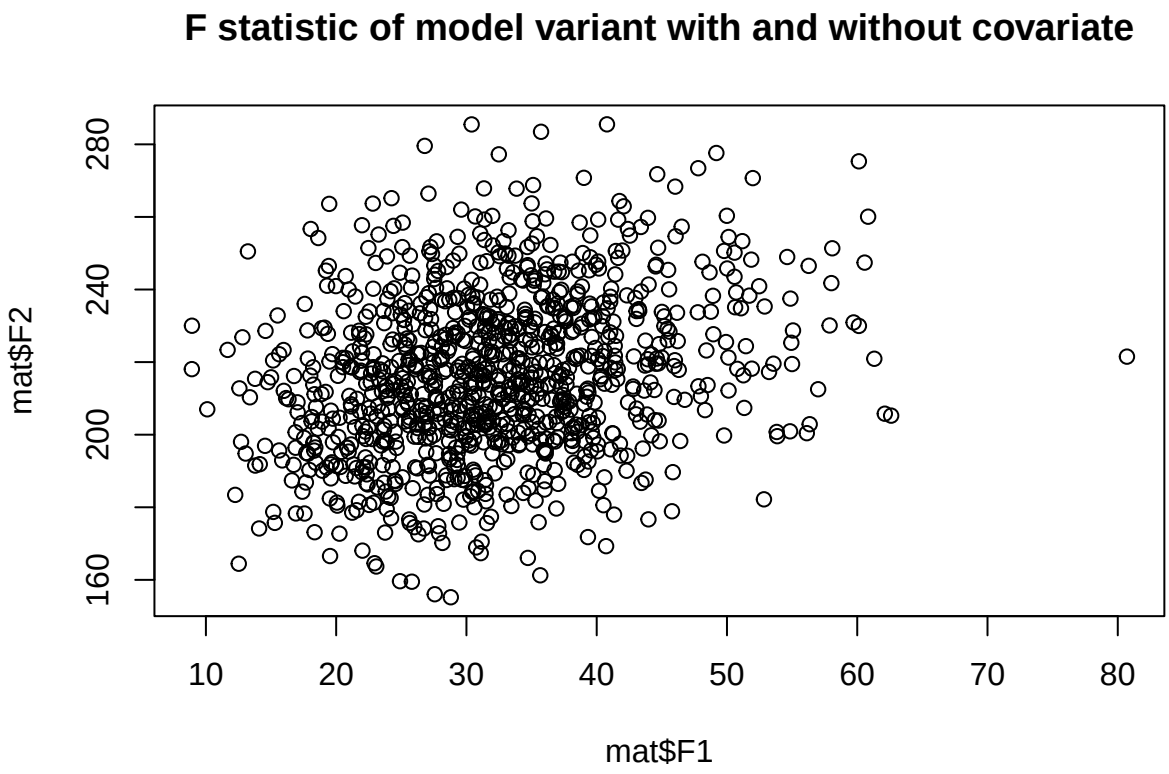

```r
mean(mat$p1-mat$p2)
```

```
## [1] 1.213308e-05
```

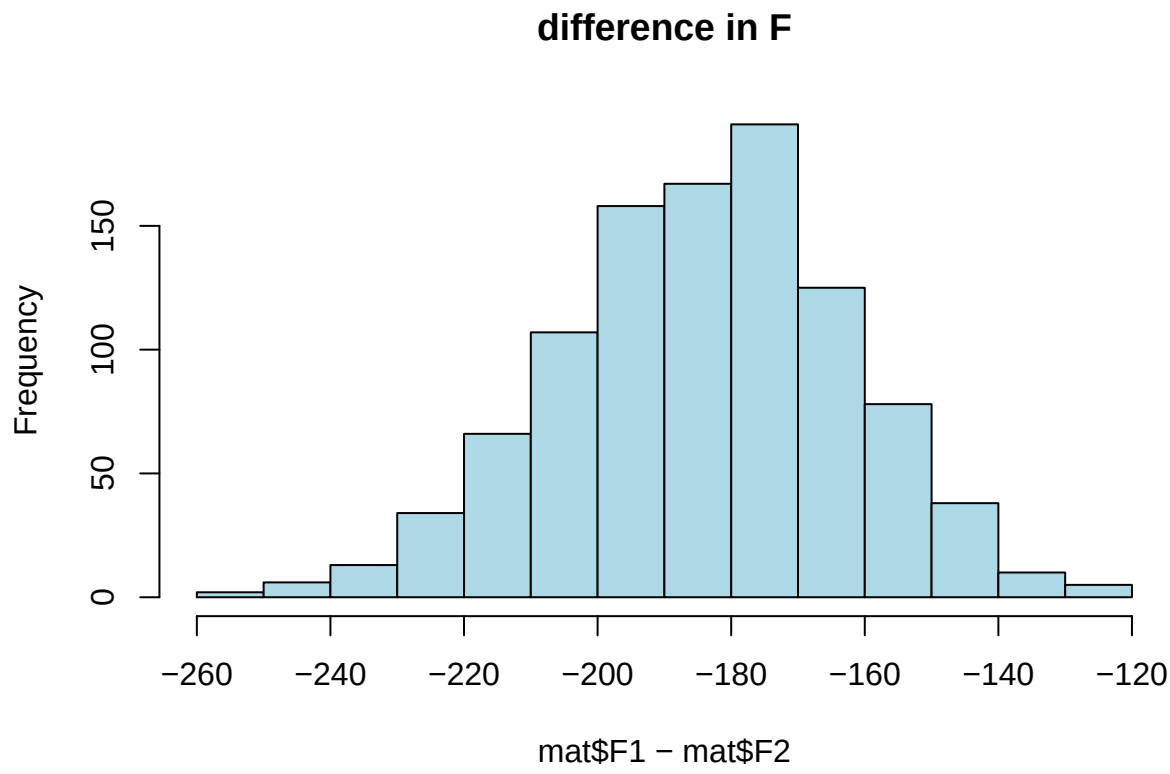
```r
max(mat$p1-mat$p2)
```

```
## [1] 0.002201136
```

```r
plot(mat$F1,mat$F2,main="F statistic of model variant with and without covariate")
abline(0,1)
```




```r
hist(mat$F1-mat$F2,breaks=10,col="light blue",main="difference in F")
```



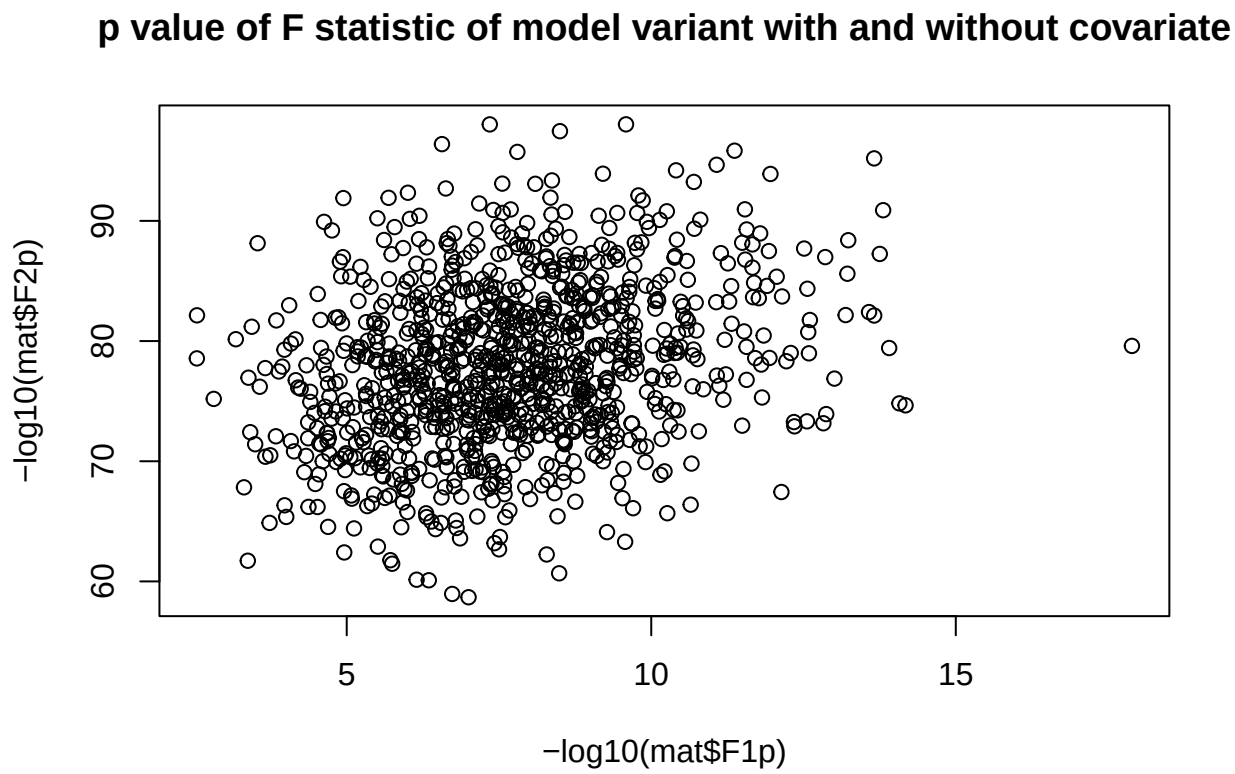

```
mean(mat$F1-mat$F2)
```

```
## [1] -184.2115
```

```
max(mat$F1-mat$F2)
```

```
## [1] -125.5856
```

```
plot(-log10(mat$F1p),-log10(mat$F2p),main="p value of F statistic of model variant with and without cova
abline(0,1)
```



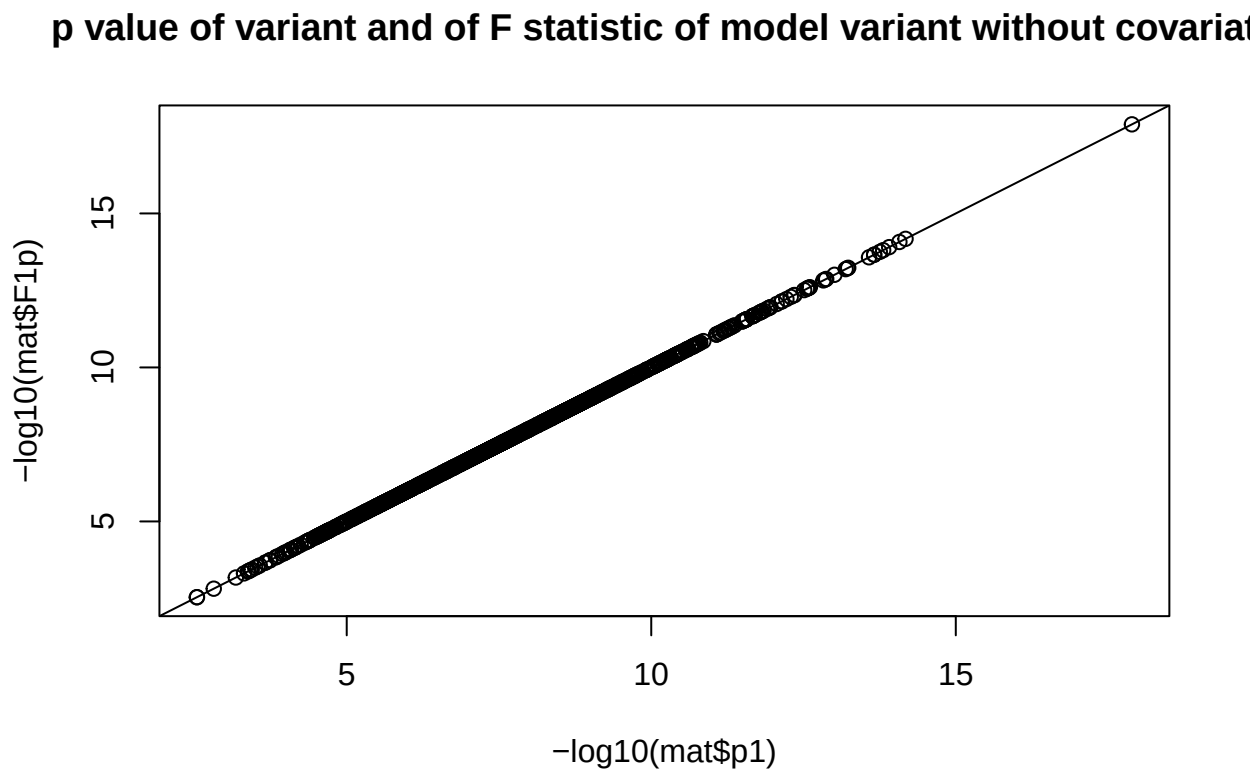

```r
mean(mat$F1p-mat$F2p)
```

```
## [1] 1.485177e-05
```

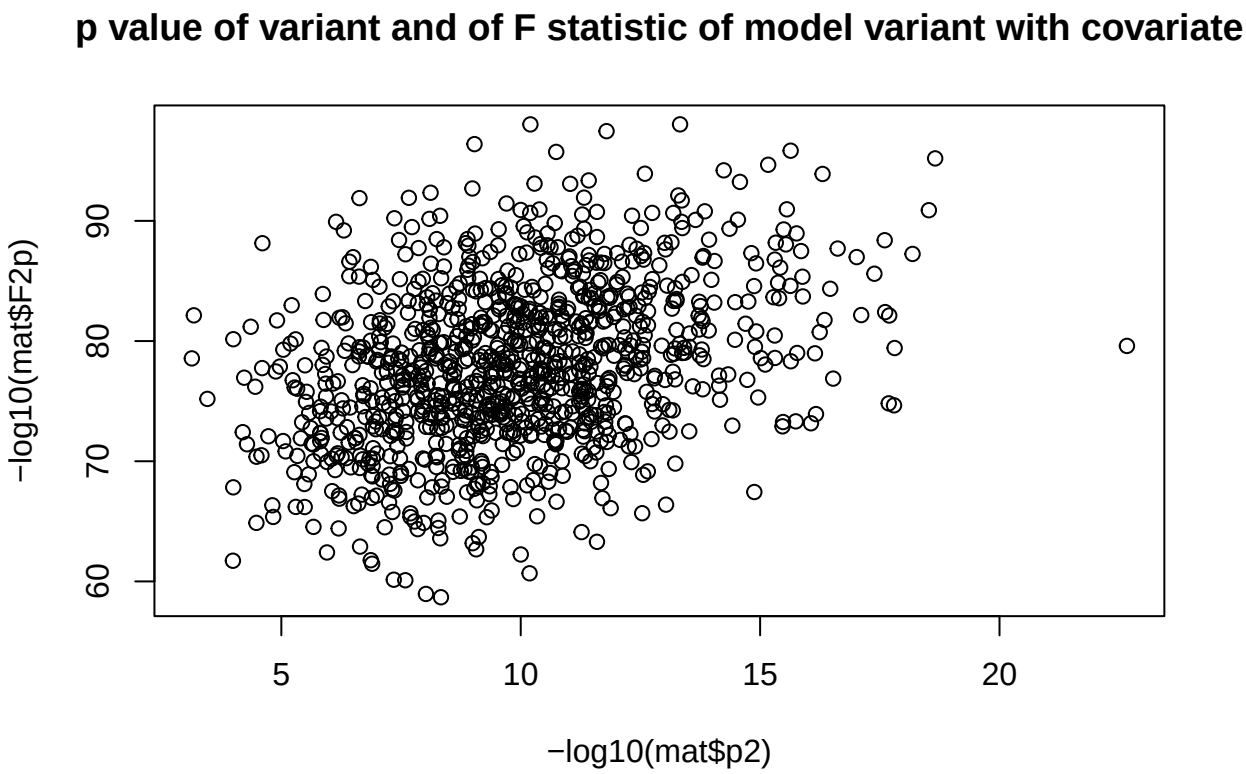
```r
max(mat$F1p-mat$F2p)
```

```
## [1] 0.002888913
```

```r
plot(-log10(mat$p1),-log10(mat$F1p),main="p value of variant and of F statistic of model variant withou
abline(0,1)
```




```
plot(-log10(mat$p2),-log10(mat$F2p),main="p value of variant and of F statistic of model variant with co
abline(0,1)
```


p value of variant and of F statistic of model variant with covariate
−log10(mat$F2p)
60
70
80
90
5
10
15
20
−log10(mat$p2)